EG2300 — Engineering Design II

Program Controller

Context

Previously you built an instruction ROM and manually selected which row to output. A processor does not manually choose instructions. It automatically moves through the program.

System overview

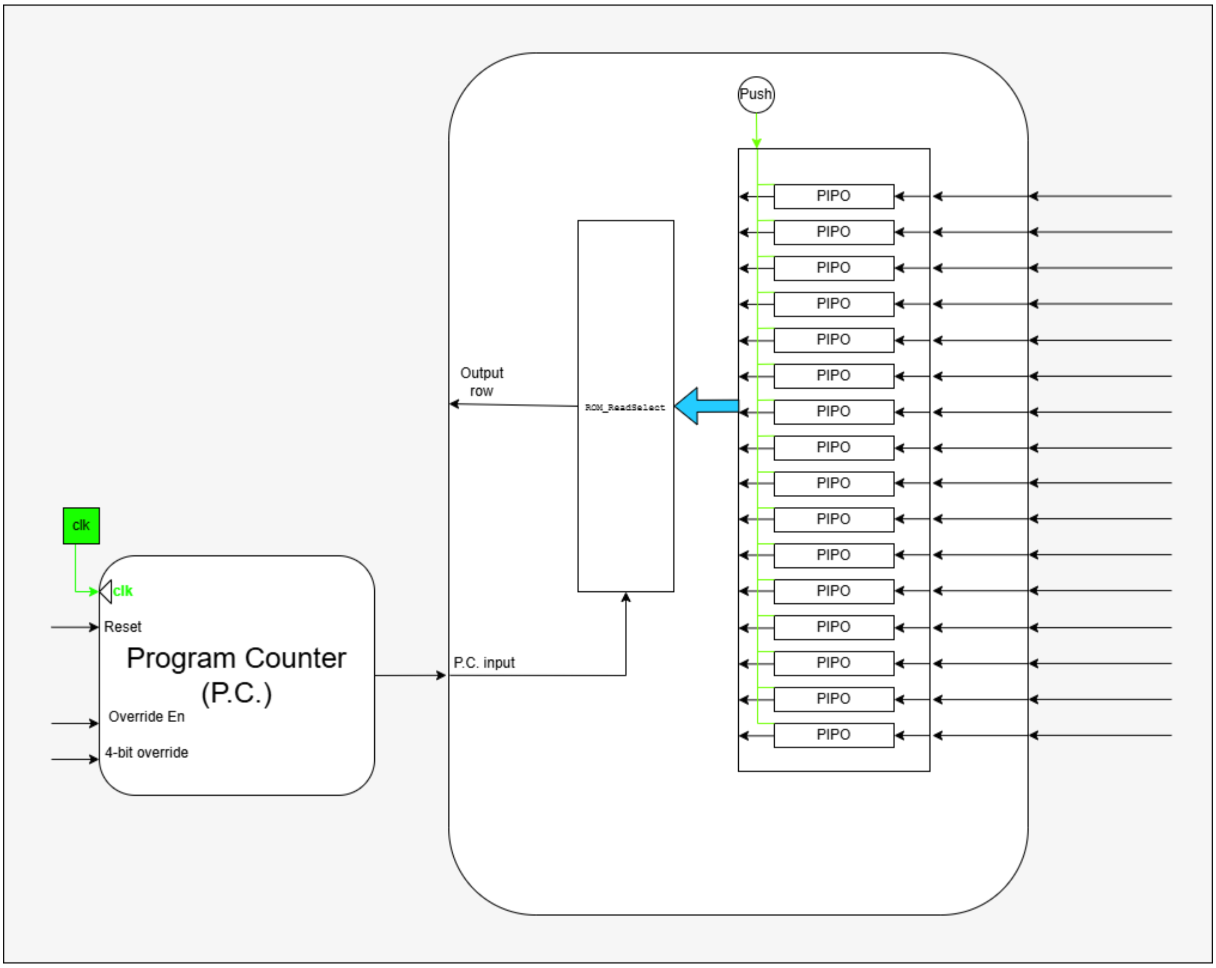

Figure 1: Instruction fetch structure. The Program Counter supplies the address of the current instruction stored in the ROM.

Goal

Build a **Program Counter (PC)** that drives the ROM address. The machine should now continuously fetch instructions.

Required behaviour

The PC is a 4-bit register that:

- Outputs the current instruction address
- Increments every clock cycle
- Resets to 0 on reboot
- Can optionally be forced to a chosen value

Two modes must exist:

Normal mode:	PC $\leftarrow$ PC + 1
Override mode:	PC $\leftarrow$ external 4-bit input

Subcircuit

Create circuit *Program_Counter* with:

Inputs

- CLOCK
- RESET
- OVERRIDE_EN
- OVERRIDE_VALUE[3:0]

Output

- PC[3:0]

Connect to ROM

Remove the manual selector from Tutorial 13. Connect PC to Instruction Address input of the ROM. Then run the clock and observe the instruction changing automatically with the clock.

Test

- PC counts 0 $\rightarrow$ 15 $\rightarrow$ 0 $\rightarrow$ 1 $\rightarrow$ ...
- Reset returns PC to 0
- When override is enabled, PC jumps to the chosen value

You now have a system that repeatedly fetches instructions, and can also jump to a chosen location. Next you will decide what those instructions actually mean.